Respiratory Alkalosis

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Adapted from: Respiratory Alkalosis

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of respiratory alkalosis.
2. Explain the pathophysiology of respiratory alkalosis.
3. Interpret clinical and lab findings to properly diagnose respiratory alkalosis.
4. Outline the management plan for respiratory alkalosis.

CASE CONNECTION

HG is a 27-year-old man who presents to the emergency department with a sudden onset of shortness of breath, palpitations, and dizziness after a stressful meeting at work. He is healthy with no significant past medical history. On exam, he is anxious, with a respiratory rate of 32 breaths/min and an oxygen saturation of 99% on room air. Basic labs as well as an arterial blood gas are obtained.

What is the likely diagnosis? What would you expect HG's blood gas results to show and why? Consider these questions as you read, and we'll revisit HG at the end of the brick.

GO TO CONCLUSION

What Is Respiratory Alkalosis?

The body depends on the pH being within a normal range in order for everything to function appropriately. The body keeps pH within a narrow range of 7.35 to 7.45. When the pH falls outside that range, the proteins or machinery of the body change their tertiary structure and function less effectively, leading to a host of problems. The natural tendency of the body is to become acidic. The reason for this is that the Krebs cycle naturally produces carbon dioxide, which gets converted into carbonic acid. The Krebs cycle produces more than 15,000 millimoles of CO_2-derived acid per day. These are called volatile acids. Non-volatile acids are acids unrelated to CO_2. These accumulate as well and have to be excreted. The body maintains acid-base balance by:

- Exhaling CO_2 through the lungs
- Excreting non-volatile acids through the kidneys

The body's pH is tightly regulated between **7.35 and 7.45**, this narrow range is essential for normal cellular function. Even small deviations can alter protein structure, enzyme activity, and cellular signaling. Severe pH shifts can cause arrhythmias, decreased cardiac contractility, neurologic impairment, and multi-organ failure.

Our body tightly controls the excretion of carbon dioxide. Respiratory alkalosis occurs when the lungs excrete excessive carbon dioxide due to rapid or deep breathing (hyperventilation). As CO_2 levels fall, less carbonic acid is available in the blood, which drives the pH upward into the alkalotic range. This disturbance can happen in situations such as anxiety, pain, sepsis, or high altitude. In each case, the underlying problem is the same: excessive loss of CO_2 through the lungs, leading to an imbalance in the body's tightly regulated acid–base system.

What is the Clinical Presentation of Respiratory Alkalosis (LO1)

Since respiratory alkalosis arises from **hyperventilation**, the dominant symptom is **rapid breathing**. When patients notice they have tachypnea, they may tell you that they are short of breath. **Dizziness**, **light-headedness**, and **paresthesias** (tingling in the fingers and around the mouth) can occur as alkalemia causes cerebral vasoconstriction and shifts calcium from its ionized to albumin-bound form, decreasing ionized calcium and thus promoting neuromuscular excitability. Sometimes, patients will present with symptoms of the underlying cause of tachypnea—cough, fever, edema, or leg swelling, which can occur with pneumonia, heart failure, and pulmonary embolism, respectively. The patient's mental state, vital signs, and context guide the differential diagnosis.

Fast breathing and dyspnea are the initial symptoms in a patient with respiratory alkalosis.

What Is the Pathophysiology of Respiratory Alkalosis? (LO2)

The body's pH is tightly regulated between **7.35 and 7.45**, a narrow range essential for normal cellular function. From a chemical perspective, pH reflects the concentration of hydrogen ions ($[H^+]$) in body fluids. The relationship between pH and $[H^+]$ is logarithmic, so each 0.3 change in pH corresponds to a doubling or halving of $[H^+]$. To manage keep the pH within a narrow range, the body uses two main systems:

Volatile acids (mainly carbon dioxide, CO_2)

- produced continuously by the Kreb's cycle in aerobic metabolism.
- CO_2 combines with water to form carbonic acid (H_2CO_3), which dissociates into hydrogen ions (H^+) and bicarbonate (HCO_3^-).
- CO_2 is eliminated by the lungs through ventilation.

Non-volatile acids (e.g., sulfuric, phosphoric, lactic acids)

- Generated from protein metabolism and other metabolic processes.

- Cannot be excreted via the lungs, so they are removed by the kidneys, which secrete H^+ and reabsorb/regenerate HCO_3^-.

CLINICAL CORRELATION

Patients with septic shock experience impaired tissue perfusion, which forces cells into anaerobic metabolism. This causes excess lactic acid production, a non-volatile acid. This lowers blood pH, but the body compensates by increasing ventilation to exhale more CO_2, which helps buffer the acidosis. At the bedside, this often presents as Kussmaul respirations, which are deep, labored breathing seen in severe metabolic acidosis.

During rapid fluctuations in acid levels throughout the day, the body needs a system to quickly neutralize acids to keep the pH stable. The **bicarbonate buffer system** is the body's primary extracellular buffer and is represented by:

$$CO_2 + H_2O \overset{CA}{\longleftrightarrow} H_2CO_3 \longleftrightarrow HCO_3^- + H^+$$

Changes in ventilation alter CO_2 levels, which shift this equation and the body's pH.

- **Hypoventilation** → ↑ CO_2 → ↑ H^+ → ↓ pH (respiratory acidosis)

- **Hyperventilation** → ↓ CO_2 → ↓ H^+ → ↑ pH (respiratory alkalosis)

In normal physiology, ventilation matches metabolic CO_2 production to keep $PaCO_2$ near 40 mmHg. Respiratory alkalosis occurs when alveolar ventilation exceeds metabolic demand, causing **excessive CO_2 excretion** (hypocapnia) and a **rise in pH**. The disturbance originates in the respiratory system, and compensation occurs through the kidneys.

Respiratory alkalosis is defined by reduced $PaCO_2$ and elevated pH.

Etiology

Anything that causes an individual to breathe faster, whether it be tachypnea to try to increase oxygen levels in the blood during pneumonia or due to inappropriate hyperventilation from anxiety, pain, hypoxemia, or central nervous system stimulation.

The causes can threrefore be subdivided into conditions due to:

- Hypoxemia (low partial pressure of oxygen [PO_2]), which causes reflex hyperventilation
- Conditions that cause hyperventilation due to other stimuli (eg, drugs, CNS)

Hypoxemic Hyperventilation. Here the low PO_2 (hypoxemia) is sensed by the peripheral chemoreceptors in the carotid body and aortic body. These signal the brain to increase the breathing rate and tidal volume, **increasing minute ventilation**. Some main causes of hypoxemia are:

- High altitude, where inspired oxygen pressure falls due to low atmospheric pressure
- Cardiopulmonary diseases (eg, congestive heart failure, pneumonia, acute respiratory distress syndrome [ARDS]), where oxygen diffusion across the alveolar membrane is reduced, leading to low PaO_2 and low oxygen delivery to tissues.

CLINICAL CORRELATION

Patients with heart failure can develop pulmonary edema, which causes a V/Q mismatch in the lungs and results in hypoxemia and hyperventilation. To treat this, patients are giving loop diuretics to resolve the pulmonary edema.

Nonhypoxic Hyperventilation. Here the hyperventilation is triggered by something other than hypoxemia (eg, pain, CNS stimuli, or systemic illness):

- CNS stimulation (eg, panic attacks, stress, pain, fear): panic disorder is the most common cause overall
- Increased metabolic demand: hyperthyroidism, sepsis, nonpulmonary infections
- Increased progestins: pregnancy—this causes bronchodilation and increased tidal volumes
- Altered serum biochemistry/hormones: cirrhosis
- Aspirin overdose: this typically causes an isolated respiratory alkalosis in early stages, then a mixed respiratory alkalosis-metabolic acidosis (due to lactic acid) later on.

CLINICAL CORRELATION

It's critical to consider pulmonary embolism (PE) as a unique but important cause of hyperventilation, which can present with hyperventilation with or without hypoxemia. The patient may or may not have chest pain. PE is potentially fatal and should be evaluated for in persons with sudden unexplained respiratory alkalosis, especially if other causes are ruled out.

Figure 1 summarizes the causes of respiratory alkalosis based on mechanism.

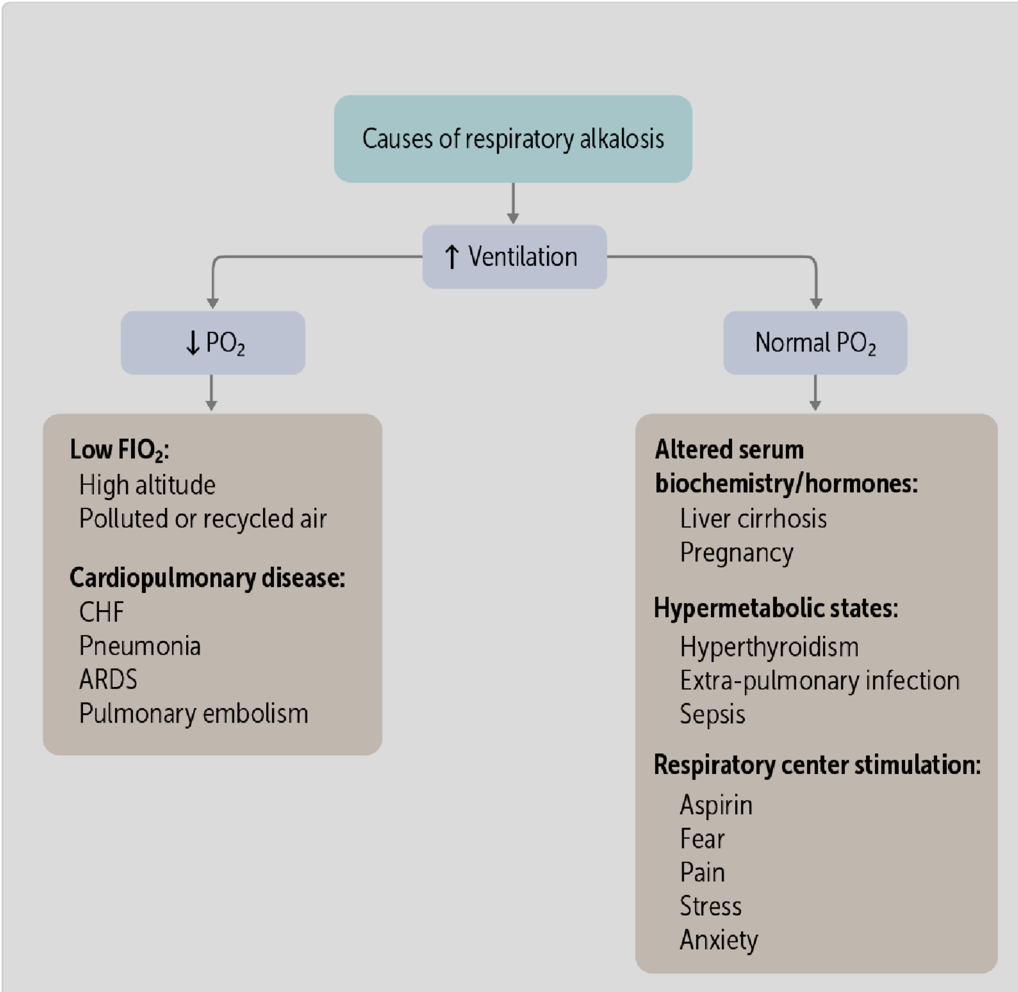

Figure 1 Causes of respiratory alkalosis

Compensation for Respiratory Alkalosis

We can summarize respiratory alkalosis as follows:

pH ↑ $PaCO_2$↓ HCO_3^-↓

Note that the $PaCO_2$ decreases in chronic respiratory alkalosis, and low $PaCO_2$ is the driver. But why is the serum bicarbonate also low? This is known as **metabolic** (**renal**) **compensation**. Here, the kidneys respond to the high blood pH by excreting bicarbonate. This compensation lowers the blood pH, bringing it back toward normal (it never actually gets all the way to normal). This compensation will take hours to days in order to fully compensate for the primary disorder, so is only noted in chronic respiratory alkalosis.

In contrast, with **acute** respiratory alkalosis (eg, due to pain or an acute pulmonary embolus) there will be no time for this compensation, so the serum bicarbonate concentration may be normal.

High altitude and cardiopulmonary diseases are causes of hypoxemic respiratory alkalosis.

How Do We Diagnose Respiratory Alkalosis? (LO3)

When a patient comes in with rapid breathing, consider respiratory alkalosis, especially in patients with known predisposing conditions, such as those who are pregnant, have leg swelling suggestive of congestive heart failure or pulmonary embolism, or those who have jaundice, which may be suggestive of liver failure. Figure 1 shows that the diagnostic algorithm starts with normal or low oxygenation. If there is a decrease in PaO_2, further pulmonary testing is warranted, beginning with imaging studies.

Laboratory Tests

A diagnosis of respiratory alkalosis is made based on venous or arterial blood gas testing and a basic metabolic panel showing the pattern of increased pH, decreased $PaCO_2$, and decreased bicarbonate (the latter seen in chronic respiratory alkalosis). Note that some hyperventilating patients are actually compensating for a metabolic acidosis (as in diabetic ketoacidosis) with deep respirations. The blood pH will easily distinguish these disorders.

Other ancillary tests, such as urine pregnancy test or thyroid function tests, may be appropriate depending on the suspected cause.

Imaging

In many cases, a thorough history and physical exam will be supplemented by a chest X-ray to evaluate for lung disease (eg, pneumonia) or congestive heart failure.

An important cause of respiratory alkalosis to always consider is pulmonary embolism, which is a potentially fatal condition that can present with or without hypoxemia. Patients may have evidence of deep venous thrombosis (eg, unilateral calf swelling). Depending on the level of suspicion, blood D-dimer testing, followed by CT angiography (CTA), can show thrombus within the pulmonary arteries to confirm pulmonary embolism (arrows in Figure 2).

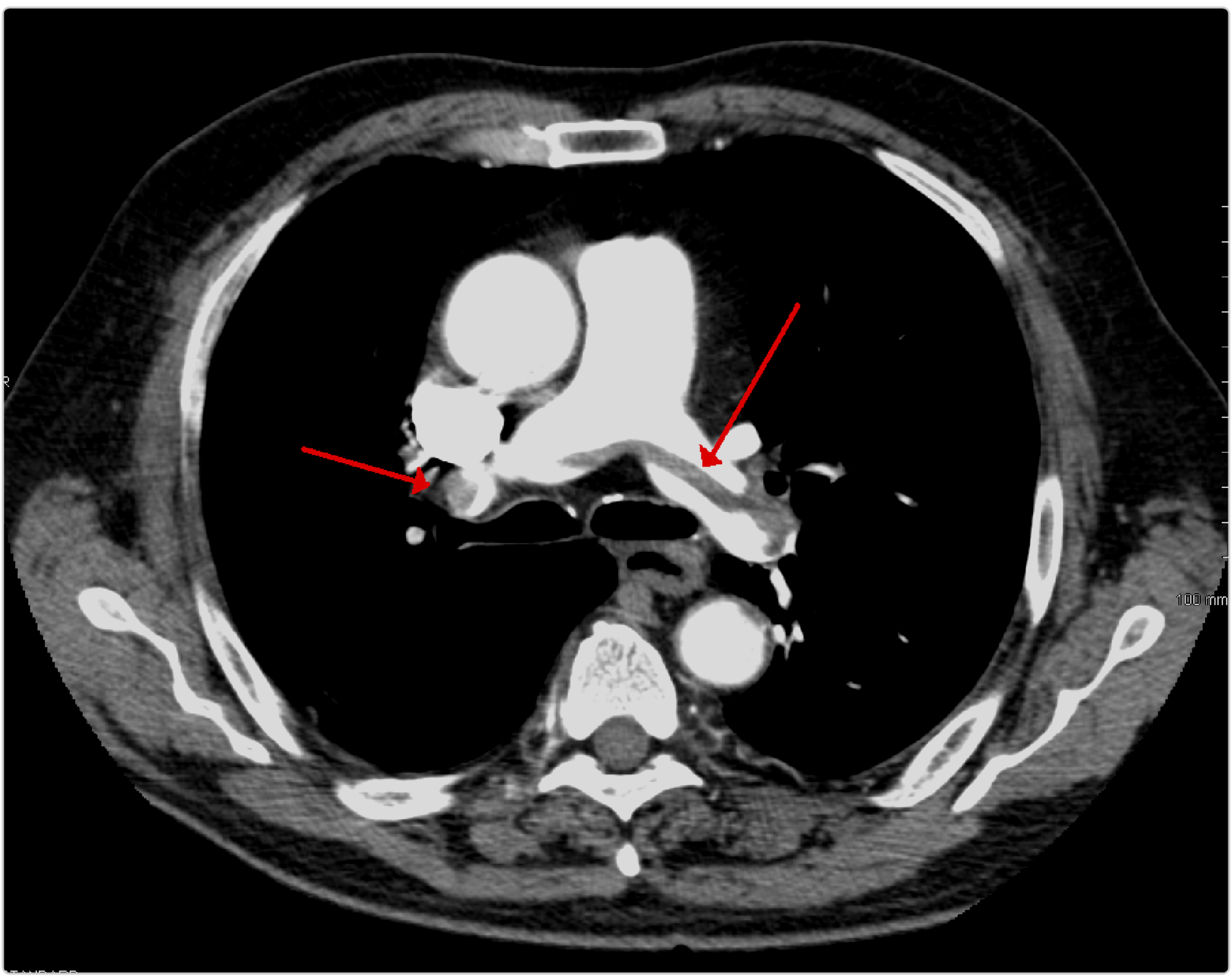

Figure 2 CT angiography in pulmonary embolism

Initially, a venous blood gas and pulse oximetry will make the diagnosis and classify the patient as hypoxemic or nonhypoxemic respiratory alkalosis.

How Do We Manage Respiratory Alkalosis? (LO4)

The overall framework for management of respiratory alkalosis is finding and treating the underlying cause. Respiratory alkalosis itself can cause symptoms from hypocalcemia due to a decrease in ionized calcium. Still, the underlying cause is the primary driver of most of their symptoms, morbidity, and mortality. Respiratory alkalosis can therefore be thought of more as a symptom and an intermediate step in the diagnosis and management process of the myriad of underlying causes. In sum, after diagnosing respiratory alkalosis, we look at oxygenation next. If there is hypoxemia or suspected cardiopulmonary disease, the overall goal is to **improve oxygenation** by administering oxygen and treating the underlying cause.

Pulmonary embolism always should be considered in patients with respiratory alkalosis because it can be fatal if untreated.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to HG, what is the likely diagnosis? What would you expect HG's blood gas results to show and why?

With only the history, the patient's rapid breathing, and good oxygen saturation of 99% on room air, you are concerned about respiratory alkalosis, possibly due to panic disorder. As expected, his arterial blood gas confirms your initial diagnosis: HG has an elevated pH and a low pCO_2. His pO_2 is normal. With reassurance and observation, HG gradually resumes a normal respiratory rate and tells you that he has had similar panic episodes in the past. He feels normal when he leaves the emergency department.

Summary

- Respiratory alkalosis is a primary acid-base disorder characterized by reduced partial pressure of arterial carbon dioxide ($PaCO_2$) and elevated blood pH.

- A high respiratory rate and dyspnea are often seen initially in respiratory alkalosis.
- Some patients will have other symptoms of cardiopulmonary disease such as cough or edema.
- Respiratory alkalosis only has one direct cause: increased alveolar ventilation.
- The cause of the hyperventilation may be hypoxemia or nonhypoxemic, such as anxiety, pain, or pulmonary embolus.
- The kidneys chronically compensate for chronic respiratory alkalosis by excreting bicarbonate, lowering the blood pH back toward normal.
- The diagnosis of respiratory alkalosis may be obvious by history, but can be confirmed by blood gas analysis.
- Pulse oximetry testing is essential to rule out hypoxemia as the cause of the hyperventilation.
- Treatment of respiratory alkalosis depends on the cause, but all hypoxic patients should get immediate oxygen.

Review Questions

1. **A 63-year-old male presents to the outpatient clinic because of increased work of breathing. He describes difficulty catching his breath over the past 6 months. He describes a 30 lbs weight gain over the past 6 months, as well as shortness of breath when he lies flat. The respiratory rate is 26 breaths/min. There are crackles in his lungs. JVD is present as is 3+ pitting lower extremity edema. His arterial blood gas shows: pH 7.55; partial pressure of arterial carbon dioxide, 25 mm Hg; bicarbonate, 21 mEq/L; partial pressure of oxygen, 70 mm Hg; and O_2 saturation, 85%. Which of the following is the most likely diagnosis?**

A. Anxiety disorder

B. Heart failure

C. Opioid overdose

D. Pulmonary embolism

Explanation (requires correct answer)

2. Which of the following conditions is most likely to cause respiratory alkalosis with a normal partial pressure of arterial oxygen?

A. Bronchitis

B. Congestive heart failure

C. High altitude

D. Hyperthyroidism

E. Pneumonia

Explanation (requires correct answer)

3. Which of the following best describes the renal response to chronic respiratory alkalosis?

A. Decreased production of H^+

B. Increased excretion of bicarbonate

C. Increased excretion of H^+

D. Increased production of bicarbonate

E. Increased reabsorption of bicarbonate

Explanation (requires correct answer)

4. **A patient presents to the emergency department with shortness of breath and tachypnea. On physical exam, they appear distressed and have rapid breathing. Pulse oximetry shows an oxygen saturation of 89% on room air. An arterial blood gas and basic metabolic panel shows respiratory alkalosis with hypoxemia. A CT chest with contrast shows a pulmonary embolism. How is this best managed?**

A. Observation until tachypnea resolves, as respiratory alkalosis is only a symptom.

B. Oxygenation only to reduce hypoxemia which is driving the rapid breathing.

C. Oxygenation as well as anticoagulation for definitive management of the underlying condition.

D. Instructions for the patient to take deep, calm breaths to reduce respiratory rate.

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Differentiate the clinical presentation of respiratory alkalosis.**

02 **Explain the pathophysiology of respiratory alkalosis.**

03 **Interpret clinical and lab findings to properly diagnose respiratory alkalosis.**

04 Outline the management plan for respiratory alkalosis.

Objective Confidence: 0/4